INCH-POUND

MIL-O-55310/5E
15 March 1994
SUPERSEDING
MIL-O-55310/5D
4 October 1984

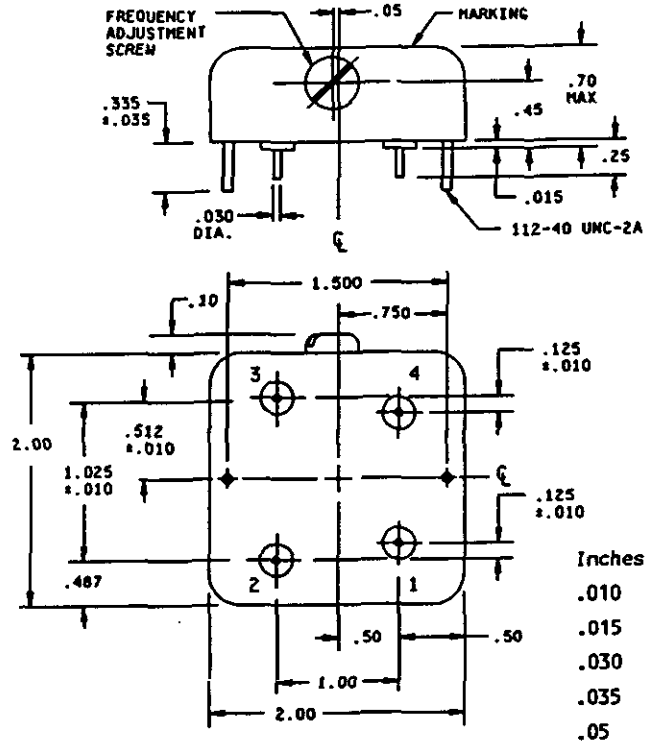
MILITARY SPECIFICATION SHEET

OSCILLATOR, CRYSTAL CONTROLLED, TYPE 3 (TEMPERATURE COMPENSATED (TCXO)), 1 THROUGH 10 MHz, SOLDER SEAL, SQUARE WAVE, TTL

Inactive for new design

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-O-55310



Configuration A
(For dash number 01)

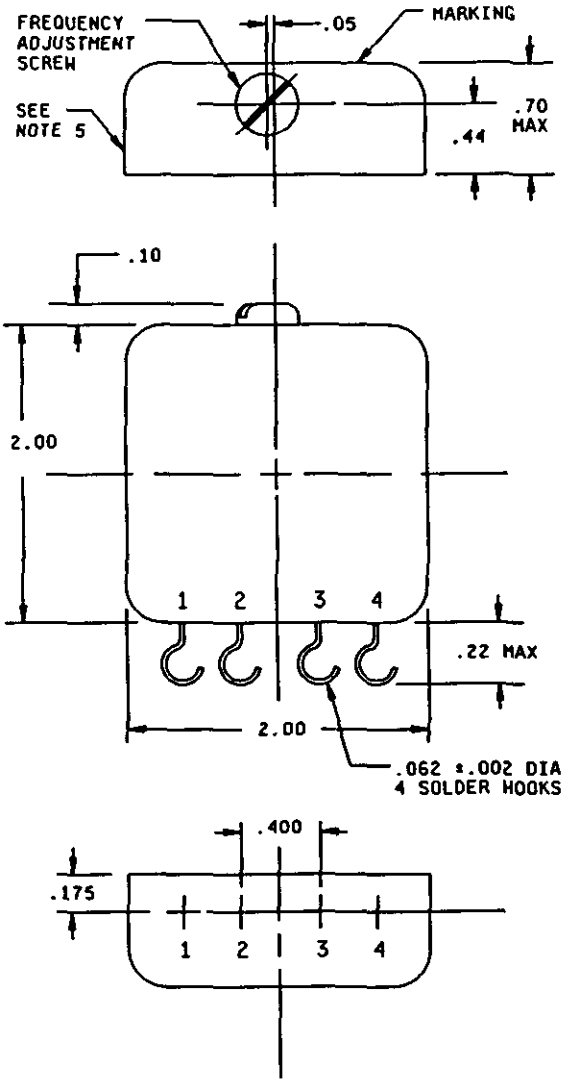
Pin number	Function
1	RF output
2	B+
3	B- (common)
4	Case

Inches	mm	Inches	mm
.010	0.25	.45	11.4
.015	0.38	.487	12.37
.030	0.76	.50	12.7
.035	0.89	.512	13.00
.05	1.3	.70	17.8
.10	2.5	.750	19.0
.112	2.84	1.00	25.4
.125	3.18	1.025	26.04
.25	6.4	1.500	38.10
.335	8.51	2.00	50.8

FIGURE 1. Dimensions and configurations.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Configuration B
(For dash number 02)

Pin number	Function
1	RF output
2	B+
3	B- (common)
4	Case

Inches	mm
.002	0.05
.05	1.3
.062	1.57
.10	2.5
.22	5.6
.44	11.2
.70	17.8

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three place decimals and ±.02 (0.5 mm) for two place decimals.
4. Normal mounting utilizes epoxy bonding to printed circuit board for configuration B only.
5. Pin number and function to be displayed on side of *oscillator*.

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS:

Design and construction: See figure 1.

Mounting: See figure 1, configurations A and B.

Terminals: See figure 1, configurations A and B.

Seal: Solder (nonhermetic).

Weight: 3 ounces maximum.

Construction technology: Discrete construction.

Components: In accordance with MIL-STD-454.

Calibration: Manufacturer and user nominal frequency calibrated.

Screening: In accordance with MIL-O-55310, class B.

Temperature:

Operating: -55°C to +85°C.

Storage: -62°C to +125°C.

Oscillator load: 1 to 10 standard TTL loads. A TTL unit load is defined as: 1.6 mA sink, 0.04 mA source and 2 pF capacitance.

Output waveform: Symmetrical square wave, TTL compatible.

Supply voltage: +5 V dc ±0.25 V dc. (+6 V dc shall not damage oscillator.)

Input current: 1 MHz through 10 MHz, 40 milliamperes maximum at 5 V dc.

Output frequency: Range shall be 1 MHz through 10 MHz; frequency as designated at time of acquisition.

Frequency adjustment (mechanical; adjustable after the cover is on): A tuning adjustment for aging shall have a range of ±5 ppm from the initial value. Resolution shall be ±0.1 ppm maximum. Grounding the tuning screw shall not damage the circuit.

Output voltage:

Logic 1: 2.4 volts minimum.

Logic 0: 0.4 volt maximum.

Rise and fall times: 15 nanoseconds maximum.

Duty cycle:

1 MHz through 4 MHz, 45 to 55 percent duty cycle at 1.4 V.

4 MHz through 10 MHz, 40 to 60 percent duty cycle at 1.4 V.

Initial frequency-temperature accuracy (one-half temperature cycle):

±5 ppm maximum from -55°C to +85°C.

±0.5 ppm maximum from +0°C to +50°C.

Measurements taken at 10 equispaced increments over the specified operating temperature range.

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Frequency-voltage tolerance: ±0.5 ppm maximum for a ±0.25 volt change in supply voltage. Measurements taken at reference temperature and operating temperature range end points.

Frequency aging: Measurements shall be taken at +70°C ±0.2°C.

±0.1 ppm per 30 days, maximum.

±0.3 ppm per 90 days, maximum. 1/

±1 ppm per year, maximum. 1/

Terminal strength: MIL-STD-202, method 211, test conditions A and B.

Applied force: 5 pounds.

Bends: Five at 45° each.

Frequency-environmental tolerance: Not applicable.

Vibration, sinusoidal: In accordance with MIL-O-55310 and MIL-STD-202, method 204.

Nonoperating: Test condition B.

Operating: Test condition C.

Ambient pressure:

Nonoperating: In accordance with MIL-O-55310.

Operating: MIL-STD-202, method 105, test condition B.

QUALITY ASSURANCE PROVISIONS

Group C inspection: Not applicable.

Part or Identifying Number (PIN): Consists of "M" prefix followed by specification sheet number, a dash and code alphas, and numeric number. See example:

EXAMPLE

M55310/05- B 01 D XXXXXXXX

M prefix and specification sheet number

Product assurance level (B only)

Dash number (see table I)

Operating temperature range (D only)

Frequency

1/ This is a performance requirement of the oscillator. Support data shall be presented showing that this requirement shall be met.

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TABLE I. Dash number and terminal location.

Dash number	Terminal location
01	See figure 1, configuration A
02	See figure 1, configuration B

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 19, 99
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5955-0677-02)

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